



INTRODUCTION TO COMPUTING

Course Outline

WEEK 1

Kick-off videos

- - https://www.youtube.com/watch?v=uMPeioU_IQs
(English)
- <https://www.youtube.com/watch?v=IP7aJd8iqAo>
(Vietnamese)

Aims

- The goal of this course is to provide **undergraduate students** with
 - basic concepts of computer science and computer engineering,
 - overview of computer science and computer engineering disciplines
 - basic programming techniques
 - With IoT devices

Outline

- fundamental concepts of computer systems
- information systems,
- computer architecture and computer hardware,
- operating systems,
- computer networks,
- computer security,
- social and ethical issues related to computing disciplines.

In particular, programming concepts including variable, assignment, branching and looping

- on IoT devices



Student learning outcomes

- L.O.1. Describe basic hardware and software concepts;
- L.O.2. Use a programming language to develop a simple project;
- L.O.3. Recognize social impacts of a computing solution to organizations and society;
- L.O.4. Demonstrate the understanding of professional, ethical, legal, and social issues and responsibilities of computing practices;



Student learning outcomes

L.O.1		Describe basic hardware and software concepts.
	L.O.1.1	Able to describe different components of a computer system such as CPU, main memory, secondary storage, input and output devices, etc.
	L.O.1.2	Able to describe different types of computer software such as system software, application software, etc.
	L.O.1.3	Able to describe basic concepts in the field of computing such as Internet, Web, Computer Networks, E-Commerce, etc.
L.O.2		Use a programming language to develop a simple project
L.O.3		Recognize social impacts of a computing solution to organizations and society
	L.O.3.1	Able to identify practical applications of a computing solution in organizations and society
	L.O.3.2	Able to give practical examples and analyze the value that computing solution can bring to organizations
L.O.4		Demonstrate the understanding of professional, ethical, legal, and social issues and responsibilities of computing practices
	L.O.4.1	Able to describe some basic job titles in Information Technology and the skills needed for these job titles
	L.O.4.2	Be aware of plagiarism and how to avoid plagiarism mistakes when writing technical reports
	L.O.4.3	Be aware of software copyright and licence
	L.O.4.4	Be aware of threats to computer systems such as virus, denial of service attacks, computer crimes, etc.
	L.O.4.5	Beware of the rapid change in computing technology, and to keep up-to-date with new technologies, computing professionals to update their knowledge continuously

Textbook/reference book

- [1] Computing Essentials: Make IT Work for , (27th edition) Timothy J. O’Leary and Linda I. O’Leary, McGraw Hill, 2019.
- [2] The C++ Programming Language, (4th Edition), Bjarne Stroustrup, Addison-Wesley, 2013

Evaluation

- Lab/exercises: 10%
- Midterm: 30% (MCQ + Additional Quizzes)
- Assignment: 30%
- Final exam: 30% (MCQ + Report)

Learning strategy

- Reading materials before the lectures
- Attending lectures => Activities on the lectures (->10%)
- Practice at the lab => They are just summaries about the lectures
- Doing projects:
 - Two assignment projects
 - Bonus: Technology Festival participation for those who performed well in the lab
- Attending final exams

Projects

- Programming projects:
 - Basic I/O
 - Loop handling

Tentative schedule

Week	Lecture	Lab	Assignment
1	Course Introduction - Flowchart		
2	Basic C++, types, variables		
3	Assignment, I/O operations		
4	Branching		
5	Switch-loop		
6	Loop (cont.) - Functions	Lab 1: Part 1 - Flowchart Part 2 - I/O C++ Part 3 - if-else	
7	JVN architecture - Basic Hardware - Basic Types	Lab 2: Part 1: Loop Part 2: Functions	Assignment 1: I/O Data Deadline Assignment 1
8	JVN architecture - Basic Hardware - Basic Types (Cont.)		Assignment 2: Loop handling Deadline Assignment 2
9	Midterm --> revision		
10	Network - IoT	IoT part 1	
11	Plagiarism --> more revision		
12	Invited Talk	IoT part 2	

Contact

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Course Website

- Course website:
 - <http://e-learning.hcmut.edu.vn>

INTRODUCTION TO ALGORITHMS AND FLOWCHARTS

- An *algorithm* is defined as a step-by-step sequence of instructions that describes how the data are to be processed to produce the desired outputs.
- In essence, an algorithm answers the question: “What method will you use to solve the problem?”.
- You can describe an algorithm by using flowchart symbols. By that way, you obtain a flowchart.
- *Flow chart* is an outline of the basic structure or logic of the program.
- Another way to describe an algorithm is using *pseudocode*.
 - English-like phrases to describe an algorithm



Flowchart Elements

Terminal



Input/output



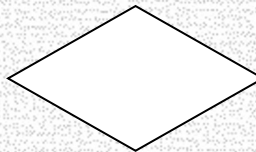
Process



Flowlines



Decision



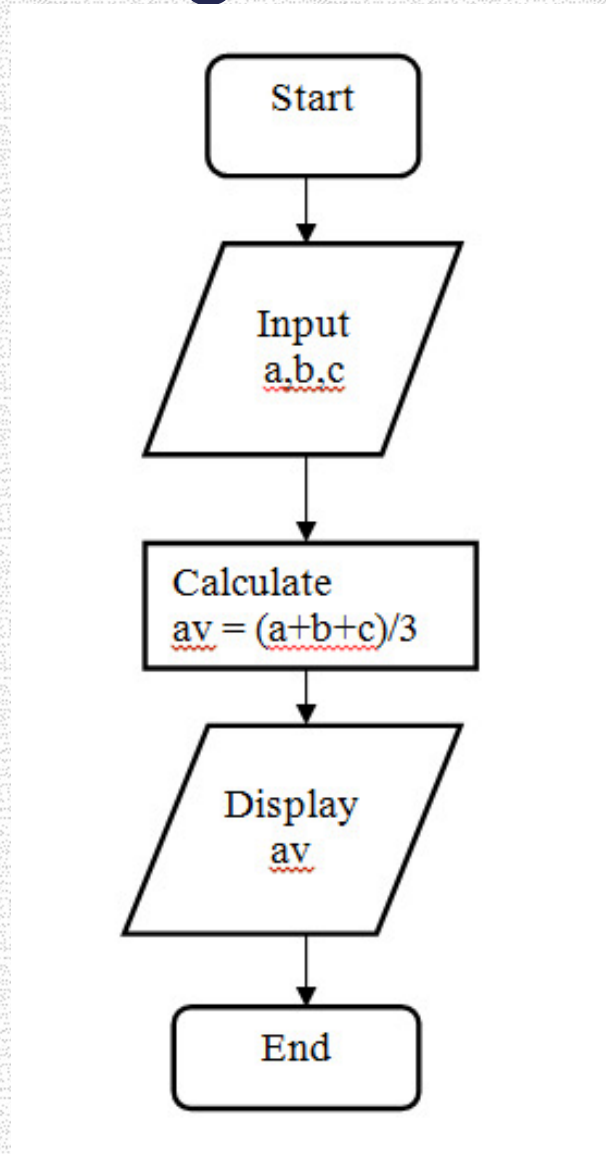
Connector



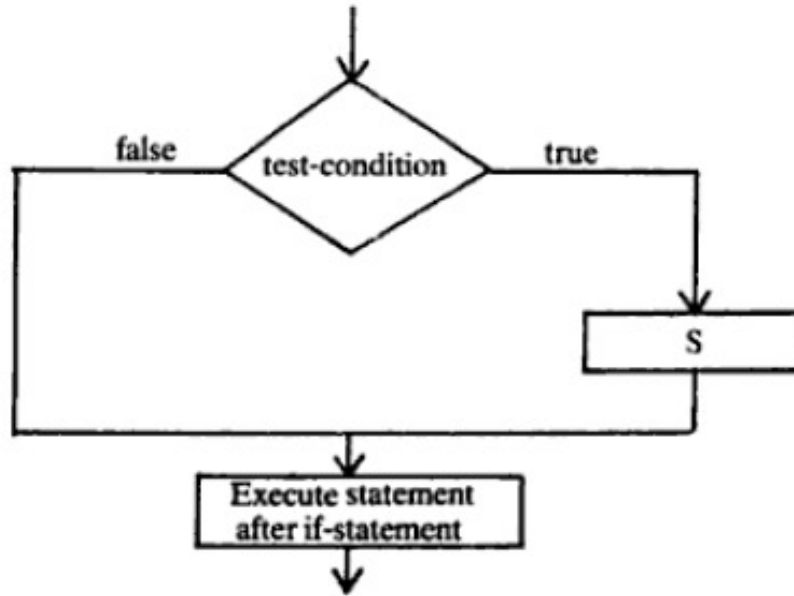
Predefined process



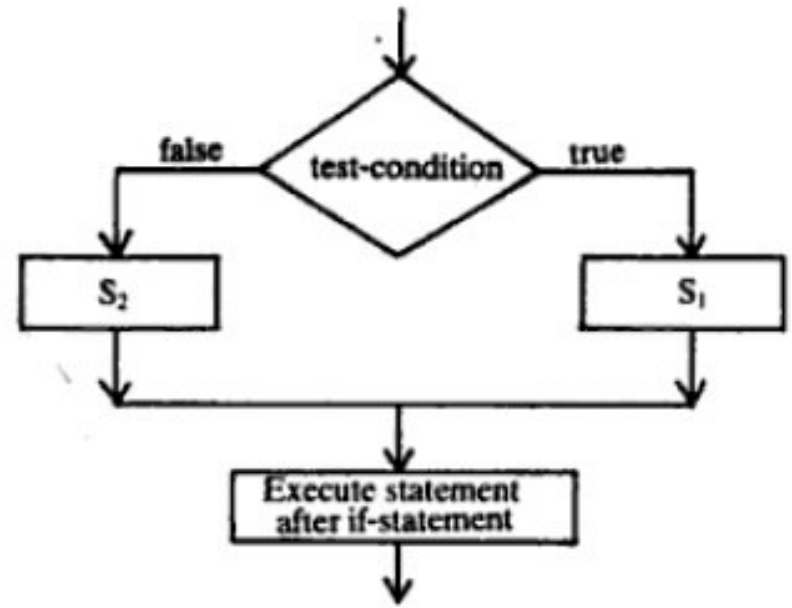
Example: Average of 3 Numbers



Branching



**If test-condition then S
endif**



**If test-condition then S₁
else S₂
endif**

Branching Example

Pseudo-code of the program:

Input A

if $A > 0$ then

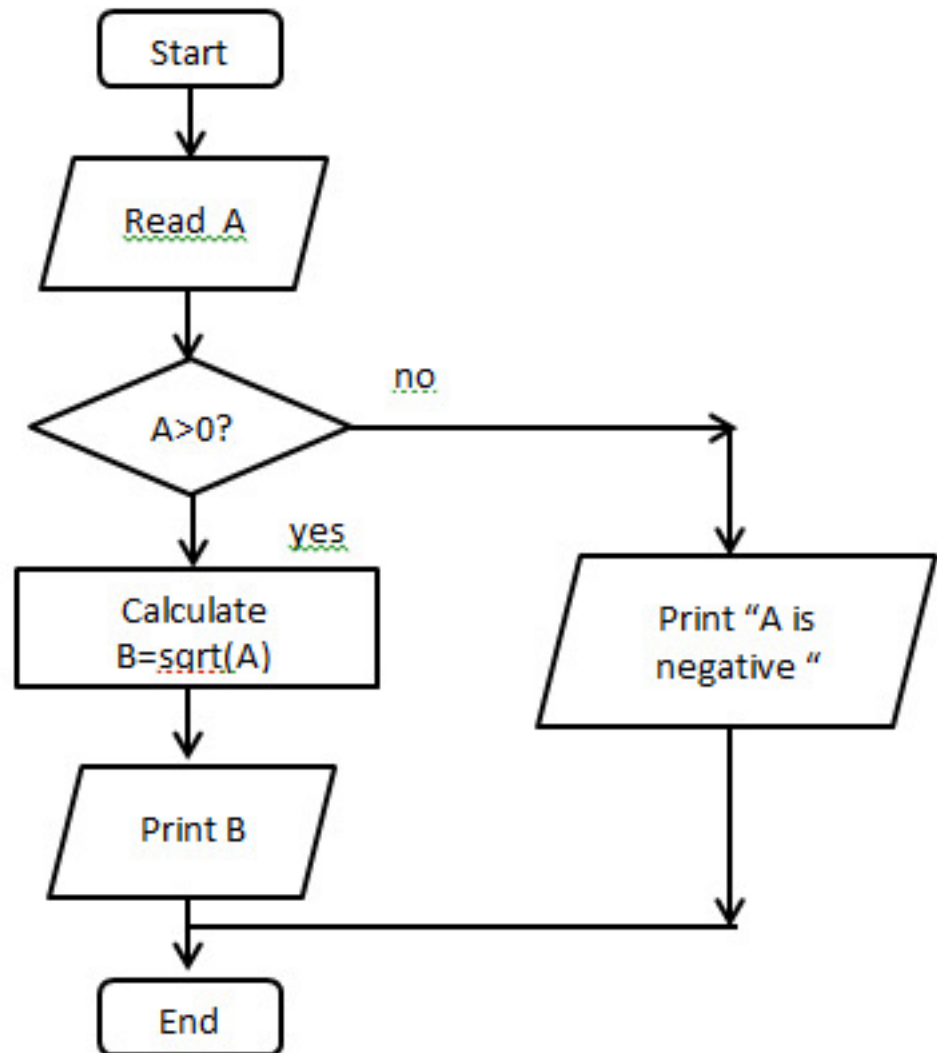
 calculate $B = \text{sqrt}(A)$

 print B

else

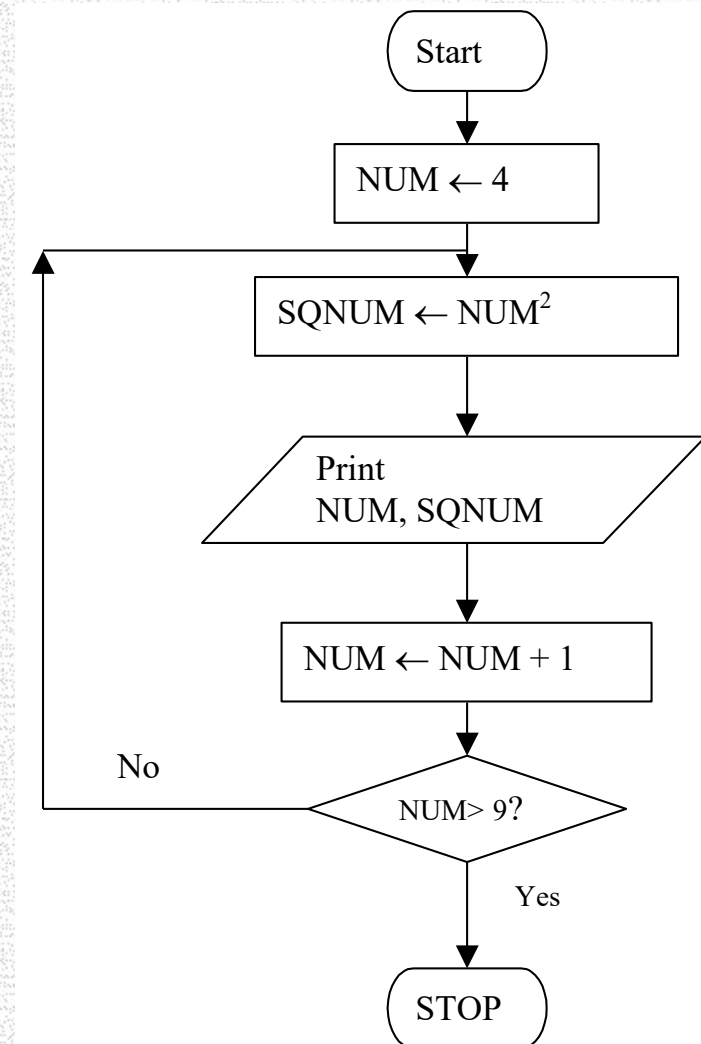
 Print "A is negative"

endif



Looping & Example

- Sequence of instructions is *repeated for as long as the loop condition is met*
- Example: write a program to do the following task.
 - Print a list of the numbers from 4 to 9, next to each number, print the square of the number.



Preparation for C++ programming

- This unit uses C++ to demonstrate many core programming concepts including branching, looping, ...
- C++ is powerful and fast
- C++ is used extensively in many areas including software, games, embedded systems development
- Install Visual Studio 2022 & Create First C++ Program | Dr Cat Can Code (English)

<https://youtu.be/MOCr-u76LiI>

- Cài đặt Visual Studio 2022 và viết chương trình C++ đầu tiên | Mèo Lập Trình (Vietnamese)

<https://youtu.be/SwpS0CIELTU>

